

Gideon A. Ong

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Education

The University of Tulsa; Tulsa, OK

Mechanical Engineering, B.S.M.E., Spanish B.A., and International Business B.S.B.A.

GPA: 4.0 | In Progress

Skills

Programming Languages: Java, JavaScript/Google Apps Script, R, Python, MATLAB, Mathematica

Technical Programs: SolidWorks, Excel, Google Sheets, PowerPoint, Scikit-learn, GIMP, Inkscape, PolyWorks

Certifications: *Black Belt – Level 4: Excel* from McGraw-Hill, CSWP, GitHub Copilot

Machine Shop: MIG Welding, CNC and Manual Mill and Lathe, Saw, Sander

Professional Skills: Clear communication, planning and coordination, adaptability, dedicated work ethic

Languages: English and Spanish (Fluent), French (Beginner)

Experience

Engineering Intern under Jason Hoines, ALS R&D Manager, at Baker Hughes, Claremore *Summer 2026*

- Automated dynamometer report data processing through pandas and a custom visualization sub-app, slashing report times of 8-40 hours by over 80%, saving upwards of \$200k annually, standardizing outputs, and empowering high-value human expertise over tedious data processing,
- Harnessed pyautogui to implement unattended 3D scanning, eliminating ~10 hours of operator idle time per batch, standardizing workflows and saving at least \$5k in labor costs annually.
- Built tools and guides for engineers to quickly and effectively implement AI to boost productivity.

Machine Shop Technician at the McElroy Prototyping Lab, University of Tulsa *Spring 2026 – present*

- Operated CNC and Manual Mills and Lathes to fabricate parts for research and student projects.

TURC Research Assistant at the University of Tulsa *Summer 2025*

- Assisted Dr. Janica Edmonds in the research and improvement of the Tulsa Math Teachers' Circle

Projects

Failure Analysis of a Titan PowrTwin 8900 Pump Cylinder *Spring 2026*

- Served as the technical lead of a 3-person team to analyze a pump cylinder failure commonly affecting Ultimate Painting & Drywall, LLC., costing \$10,000-\$15,000 per failure in repair and downtime.
- Designed, machined, tested, and installed a release pin to address a diagnosed water-freezing failure in a pump cylinder, solving a \$10,000+ per failure cost with an easy-to use \$135 mechanical solution.
- Won the 2026 Hackworth-Willson Prize for Excellence in Failure Analysis.

Chilled Water Pipeline System Design and Analysis *Fall 2025*

- Routed and sized a replacement chilled-water pipeline for TU's campus in Python, using AI-assisted convex-hull routing and a separate pipe-diameter optimization; it succeeded as the lowest-cost bid.

Finite Element Analysis Stress Relief Design for a Notched Beam in 4-Point Bending *Fall 2025*

- Leveraged SolidWorks FEA simulation to compare and optimize stress relief designs for a notched beam in bending, controlling for material variations with statistical reasoning.
- Achieved a 42% reduction in the stress concentration factor with a novel spline geometry.

Factor Analysis with Mixed Data (FAMD) on Bicycle Crashes in Scikit-learn *Spring 2025*

- Combined FAMD and undersampling as part of a preprocessing pipeline to concurrently optimize dimension reduction and class balancing for each applied model.

Spatial Analysis and Machine Learning with Car Crash Data in Scikit-learn *Spring 2025*

- Implemented DBSCAN clustering techniques on UK car crash data to identify the most accident-prone intersections by year and flag recurrent problematic intersections.
- Harnessed QGIS to effectively display the spatial relationships captured by the clustering algorithms, efficiently illustrating how DBSCAN removed noise and placed markers.

Eagle Project – Building a Wheelchair-Accessible Path *Summer 2022*

- Led 27 volunteers in 284 combined work-hours in constructing a self-designed 275.5 sq ft pathway in the backyard of The Bee Community, a business that employs adults with special needs.